

laporan Pemrograman Mobile 3



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2026

Note :

1. Setstate lebih baik digunakan untuk objek yang digunakan di 1 halaman saja, untuk penggunaan objek yang akan digunakan di lebih dari 1 halaman sebaiknya menggunakan provider.

Checkpoint A :

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 3',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const InputPage(),
    );
  }
}

class InputPage extends StatefulWidget {
  const InputPage({super.key});

  @override
  State<InputPage> createState() => _InputPageState();
}

class _InputPageState extends State<InputPage> {
  final _controller = TextEditingController();
  String _salam = '';

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

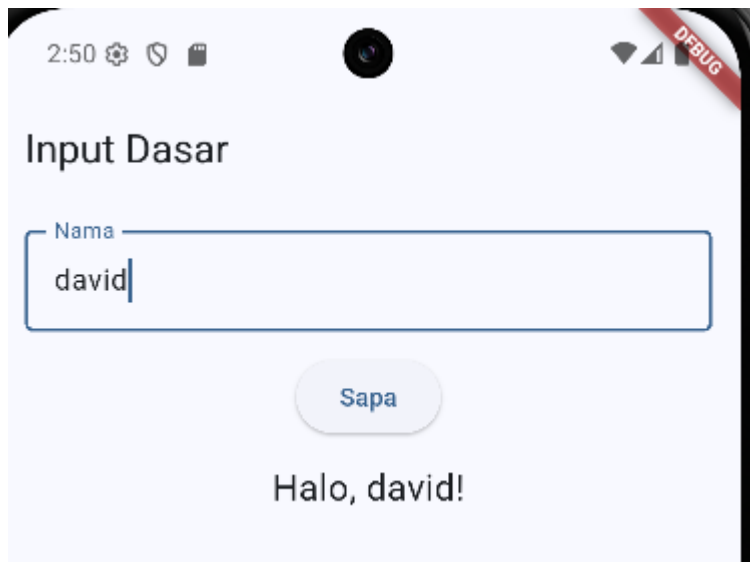
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Input Dasar'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            TextField(
              controller: _controller,
              decoration: const InputDecoration(
                labelText: 'Nama',
                border: OutlineInputBorder(),
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

```

    ),
  ),
  const SizedBox(height: 12),
  ElevatedButton(
    onPressed: () {
      setState(() => _salam = 'Halo, ${_controller.text}!');
    },
    child: const Text('Sapa'),
  ),
  const SizedBox(height: 12),
  Text(_salam, style: const TextStyle(fontSize: 20),
  ),
],
),
),
),
);
}
}

```

Hasil :



Checkpoint B :

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 3',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const FormPage(),
    );
  }
}

class FormPage extends StatefulWidget {
  const FormPage({super.key});

  @override
  State<FormPage> createState() => _FormPageState();
}

class _FormPageState extends State<FormPage> {
  final _formKey = GlobalKey<FormState>();
  final _nama = TextEditingController();
  final _email = TextEditingController();
  String? _jurusan;
  bool _setuju = false;

  @override
  void dispose() {
    _nama.dispose();
    _email.dispose();
    super.dispose();
  }

  void _ kirim() {
    if (_formKey.currentState!.validate()) {
      final jurusan = _jurusan ?? '-';
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text('Terdaftar: ${_nama.text} ($jurusan)'),
      );
    }
  }

  @override
  Widget build(BuildContext context) {
```

```

return Scaffold(
  appBar: AppBar(
    title: const Text('Form Pendaftaran'),
  ),
  body: Form(
    key: _formKey,
    child: ListView(
      padding: const EdgeInsets.all(16),
      children: [
        TextFormField(
          controller: _nama,
          decoration: const InputDecoration(
            labelText: 'Nama lengkap',
            border: OutlineInputBorder(),
          ),
          validator: (v) =>
            (v == null || v.trim().isEmpty) ? 'Nama wajib diisi' : null,
        ),
        const SizedBox(height: 12),
        TextFormField(
          controller: _email,
          keyboardType: TextInputType.emailAddress,
          decoration: const InputDecoration(
            labelText: 'Email',
            border: OutlineInputBorder(),
          ),
          validator: (v) {
            if (v == null || !v.contains('@')) return 'Email tidak valid';
            return null;
          },
        ),
        const SizedBox(height: 12),
        DropdownButtonFormField<String>(
          decoration: const InputDecoration(
            labelText: 'Jurusan',
            border: OutlineInputBorder(),
          ),
          items: const [
            DropdownMenuItem(
              value: 'TI',
              child: Text('Teknik Informatika'),
            ),
            DropdownMenuItem(
              value: 'SI',
              child: Text('Sistem Informasi'),
            ),
            DropdownMenuItem(
              value: 'TE',
              child: Text('Teknik Elektro'),
            ),
          ],
          onChanged: (v) => setState(() => _jurusan = v),
          validator: (v) => v == null ? 'Pilih jurusan' : null,
        ),
      ],
    ),
  ),
);

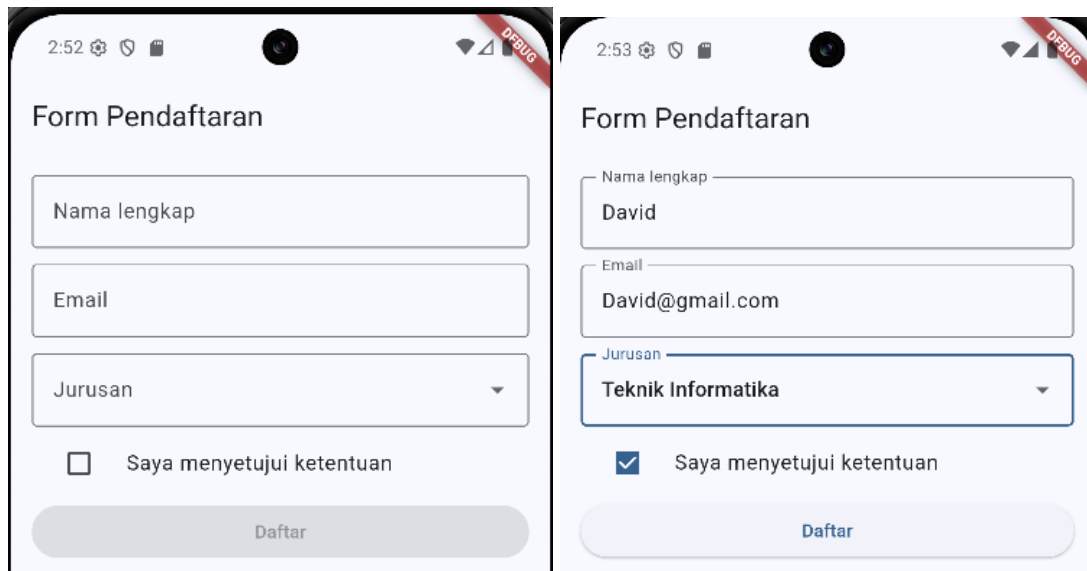
```

```

    ),
    CheckboxListTile(
      title: const Text('Saya menyetujui ketentuan'),
      value: _setuju,
      controlAffinity: ListTileControlAffinity.leading,
      onChanged: (v) => setState(() => _setuju = v ?? false),
    ),
    ElevatedButton(
      onPressed: _setuju ? _ kirim : null,
      child: const Text('Daftar'),
    ),
  ],
),
),
);
}
}

```

Hasil :



2:53

Form Pendaftaran

Nama lengkap

David

Email

David@gmail.com

Jurusan

Teknik Informatika

☒ Saya menyetujui ketentuan

Daftar

Terdaftar: David (TI)

3

Error jika gmail tanpa @ :

Email

Davidgmail.com

Email tidak valid

Checkpoint C :

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Tugas {
  String judul;
  bool selesai;

  Tugas(this.judul, {this.selesai = false});
}

class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];

  List<Tugas> get items => List.unmodifiable(_items);
  int get jumlahSelesai => _items.where((t) => t.selesai).length;

  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }

  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Daftar Tugas',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
      ),
    );
  }
}
```

```

        useMaterial3: true,
    ),
    home: const TugasPage(),
);
}
}

class TugasPage extends StatelessWidget {
  const TugasPage({super.key});

  @override
  Widget build(BuildContext context) {
    final model = context.watch<TugasModel>();

    return Scaffold(
      appBar: AppBar(
        title: Text('Tugas (${model.jumlahSelesai}/${model.items.length})'),
      ),
      body: ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final t = model.items[i];
          return ListTile(
            leading: Checkbox(
              value: t.selesai,
              onChanged: (_) => context.read<TugasModel>().toggle(i),
            ),
            title: Text(
              t.judul,
              style: TextStyle(
                decoration: t.selesai ? TextDecoration.lineThrough : null,
              ),
            ),
            trailing: IconButton(
              icon: const Icon(Icons.delete),
              onPressed: () => context.read<TugasModel>().hapus(i),
            ),
          );
        },
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          Navigator.push(
            context,
            MaterialPageRoute(
              builder: (_) => const TambahPage(),
            ),
          );
        },
        child: const Icon(Icons.add),
      ),
    );
  }
}

```

```

}

class TambahPage extends StatefulWidget {
  const TambahPage({super.key});

  @override
  State<TambahPage> createState() => _TambahPageState();
}

class _TambahPageState extends State<TambahPage> {
  final _controller = TextEditingController();

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

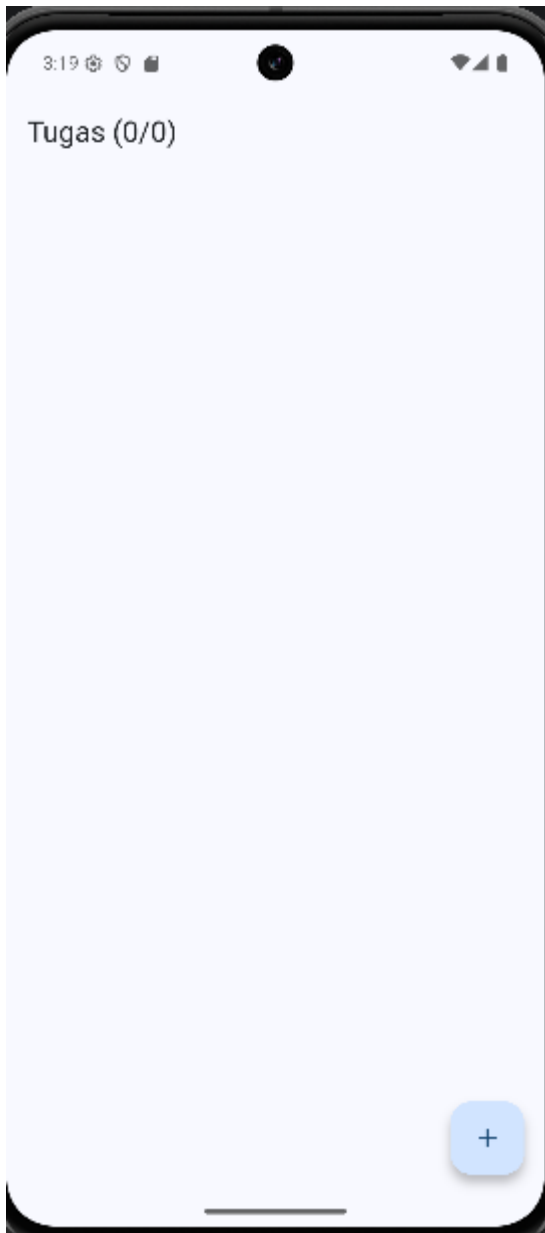
  void _simpan() {
    final judul = _controller.text.trim();
    if (judul.isEmpty) return;
    context.read<TugasModel>().tambah(judul);
    Navigator.pop(context);
  }

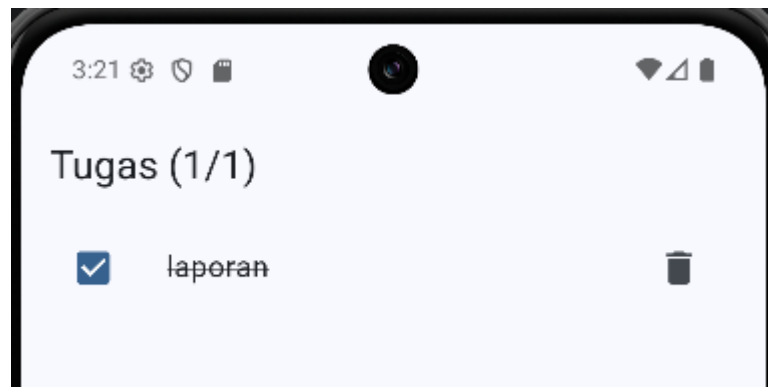
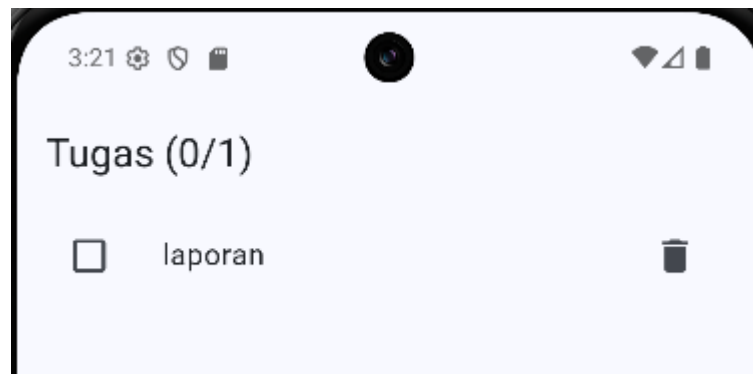
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Tambah Tugas'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            TextField(
              controller: _controller,
              autofocus: true,
              decoration: const InputDecoration(
                labelText: 'Judul tugas',
                border: OutlineInputBorder(),
              ),
              onSubmitted: (_) => _simpan(),
            ),
            const SizedBox(height: 12),
            ElevatedButton(
              onPressed: _simpan,
              child: const Text('Simpan'),
            ),
          ],
        ),
      ),
    );
  }
}

```

```
}  
}
```

Hasil :





Latihan Mandiri :

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Tugas {
  String judul;
  bool selesai;

  Tugas(this.judul, {this.selesai = false});
}

class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];

  List<Tugas> get items => List.unmodifiable(_items);
  int get jumlahSelesai => _items.where((t) => t.selesai).length;

  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }

  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }

  void hapusSelesai() {
    _items.removeWhere((t) => t.selesai);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
```

```

        debugShowCheckedModeBanner: false,
        title: 'Daftar Tugas',
        theme: ThemeData(
            colorSchemeSeed: Colors.blue,
            useMaterial3: true,
        ),
        home: const TugasPage(),
    );
}
}

class TugasPage extends StatelessWidget {
    const TugasPage({super.key});

    @override
    Widget build(BuildContext context) {
        final model = context.watch<TugasModel>();

        return Scaffold(
            appBar: AppBar(
                title: Text('Tugas (${model.jumlahSelesai}/${model.items.length})'),
                actions: [
                    IconButton(
                        icon: const Icon(Icons.delete_sweep),
                        tooltip: 'Hapus Tugas Selesai',
                        onPressed: () => context.read<TugasModel>().hapusSelesai(),
                    ),
                ],
            ),
            body: model.items.isEmpty
                ? const Center(
                    child: Text(
                        'Belum ada tugas',
                        style: TextStyle(fontSize: 18, color: Colors.grey),
                    ),
                )
                : ListView.builder(
                    itemCount: model.items.length,
                    itemBuilder: (context, i) {
                        final t = model.items[i];
                        return ListTile(
                            leading: Checkbox(
                                value: t.selesai,
                                onChanged: (_) => context.read<TugasModel>().toggle(i),
                            ),
                            title: Text(
                                t.judul,
                                style: TextStyle(
                                    decoration:
                                        t.selesai ? TextDecoration.lineThrough : null,
                                ),
                            ),
                        ),
                    ),
                ),
        );
    }
}

```

```

        trailing: IconButton(
            icon: const Icon(Icons.delete),
            onPressed: () => context.read<TugasModel>().hapus(i),
        ),
    ),
);
},
),
floatingActionButton: FloatingActionButton(
    onPressed: () {
        Navigator.push(
            context,
            MaterialPageRoute(
                builder: (_) => const TambahPage(),
            ),
        );
    },
    child: const Icon(Icons.add),
),
);
}
}

```

```

class TambahPage extends StatefulWidget {
    const TambahPage({super.key});

    @override
    State<TambahPage> createState() => _TambahPageState();
}

```

```

class _TambahPageState extends State<TambahPage> {
    final _formKey = GlobalKey<FormState>();
    final _controller = TextEditingController();

    @override
    void dispose() {
        _controller.dispose();
        super.dispose();
    }

    void _simpan() {
        if (_formKey.currentState!.validate()) {
            context.read<TugasModel>().tambah(_controller.text.trim());
            Navigator.pop(context);

            ScaffoldMessenger.of(context).showSnackBar(
                const SnackBar(content: Text('Tugas ditambahkan')),
            );
        }
    }
}

```

```

@override
Widget build(BuildContext context) {
    return Scaffold(

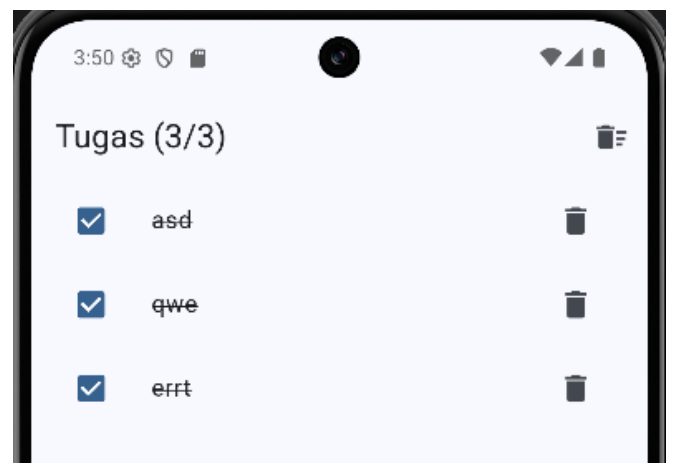
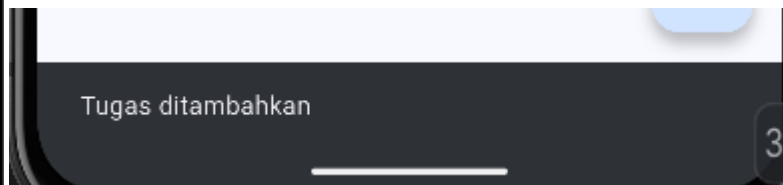
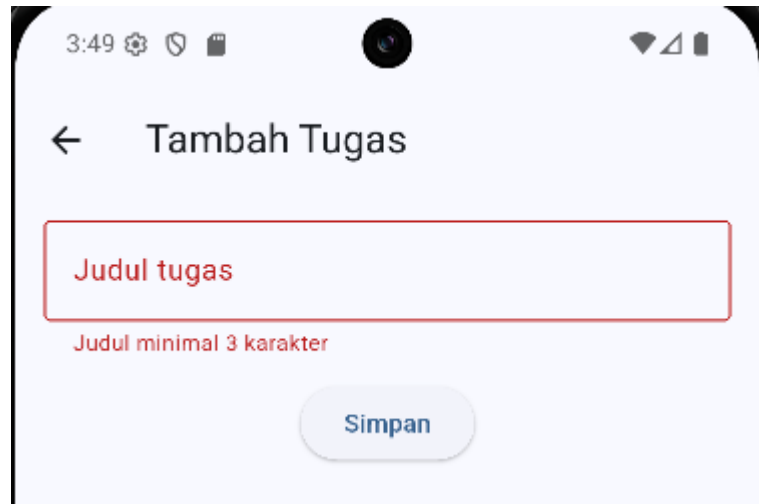
```

```

appBar: AppBar(
  title: const Text('Tambah Tugas'),
),
body: Padding(
  padding: const EdgeInsets.all(16),
  child: Form(
    key: _formKey,
    child: Column(
      children: [
        TextFormField(
          controller: _controller,
          autofocus: true,
          decoration: const InputDecoration(
            labelText: 'Judul tugas',
            border: OutlineInputBorder(),
          ),
          validator: (v) {
            if (v == null || v.trim().length < 3) {
              return 'Judul minimal 3 karakter';
            }
            return null;
          },
          onFieldSubmitted: (_) => _simpan(),
        ),
        const SizedBox(height: 12),
        ElevatedButton(
          onPressed: _simpan,
          child: const Text('Simpan'),
        ),
      ],
    ),
  ),
),
);
}
}

```

Hasil :



Tugas :

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class belanjaitem {
  String nama;
  int jumlah;
  String kategori;
  bool sudahdibeli;

  belanjaitem({
    required this.nama,
    required this.jumlah,
    required this.kategori,
    this.sudahdibeli = false,
  });
}

class belanjamodel extends ChangeNotifier {
  final List<belanjaitem> _items = [];

  List<belanjaitem> get items => List.unmodifiable(_items);

  int get jumlahBelumDibeli => _items.where((item) =>
!item.sudahdibeli).length;

  void tambah(String nama, int jumlah, String kategori) {
    _items.add(belanjaitem(
      nama: nama,
      jumlah: jumlah,
      kategori: kategori,
    ));
    notifyListeners();
  }

  void toggleStatus(int index) {
    _items[index].sudahdibeli = !_items[index].sudahdibeli;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => belanjamodel(),
      child: const MyApp(),
    ),
  );
}
```

```

}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Daftar Belanja',
      theme: ThemeData(
        colorSchemeSeed: Colors.green,
        useMaterial3: true,
      ),
      home: const DaftarBelanjaPage(),
    );
  }
}

class DaftarBelanjaPage extends StatelessWidget {
  const DaftarBelanjaPage({super.key});

  @override
  Widget build(BuildContext context) {
    final model = context.watch<belanjamodel>();

    return Scaffold(
      appBar: AppBar(
        title: Text('Belum Dibeli: ${model.jumlahBelumDibeli}'),
      ),
      body: model.items.isEmpty
        ? const Center(
            child: Text(
              'Daftar belanjaan kosong',
              style: TextStyle(fontSize: 18, color: Colors.grey),
            ),
          )
        : ListView.builder(
            itemCount: model.items.length,
            itemBuilder: (context, index) {
              final item = model.items[index];
              return ListTile(
                leading: Checkbox(
                  value: item.sudahdibeli,
                  onChanged: (_) =>
                    context.read<belanjamodel>().toggleStatus(index),
                ),
                title: Text(
                  item.nama,
                  style: TextStyle(
                    decoration: item.sudahdibeli
                      ? TextDecoration.lineThrough
                      : null,

```

```

                fontWeight: FontWeight.bold,
            ),
        ),
        subtitle: Text(
            'Jumlah: ${item.jumlah} | Kategori: ${item.kategori}',
        ),
        trailing: IconButton(
            icon: const Icon(Icons.delete, color: Colors.red),
            onPressed: () =>
                context.read<belanjaModel>().hapus(index),
        ),
    );
},
),
floatingActionButton: FloatingActionButton(
    onPressed: () {
        Navigator.push(
            context,
            MaterialPageRoute(
                builder: (_) => const FormTambahPage(),
            ),
        );
    },
    child: const Icon(Icons.add),
),
);
}
}

```

```

class FormTambahPage extends StatefulWidget {
    const FormTambahPage({super.key});

    @override
    State<FormTambahPage> createState() => _FormTambahPageState();
}

```

```

class _FormTambahPageState extends State<FormTambahPage> {
    final _formKey = GlobalKey<FormState>();
    final _namaController = TextEditingController();
    final _jumlahController = TextEditingController();
    String? _kategoriTerpilih;

    final List<String> _daftarKategori = [
        'Makanan & Minuman',
        'Kebutuhan Rumah',
        'Bumbu Dapur',
        'Elektronik',
        'Lainnya',
    ];

    @override
    void dispose() {
        _namaController.dispose();
    }
}

```

```

    _jumlahController.dispose();
    super.dispose();
}

void _simpan() {
  if (_formKey.currentState!.validate()) {
    final nama = _namaController.text.trim();
    final jumlah = int.parse(_jumlahController.text.trim());

    context.read<belanjamodel>().tambah(nama, jumlah, _kategoriTerpilih!);
    Navigator.pop(context);

    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(content: Text('Barang berhasil ditambahkan!')),
    );
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Tambah Belanjaan'),
    ),
    body: Padding(
      padding: const EdgeInsets.all(16.0),
      child: Form(
        key: _formKey,
        child: ListView(
          children: [
            TextFormField(
              controller: _namaController,
              decoration: const InputDecoration(
                labelText: 'Nama Barang',
                border: OutlineInputBorder(),
              ),
              validator: (value) {
                if (value == null || value.trim().isEmpty) {
                  return 'Nama barang wajib diisi';
                }
                return null;
              },
            ),
            const SizedBox(height: 16),
            TextFormField(
              controller: _jumlahController,
              keyboardType: TextInputType.number,
              decoration: const InputDecoration(
                labelText: 'Jumlah',
                border: OutlineInputBorder(),
              ),
              validator: (value) {

```

```

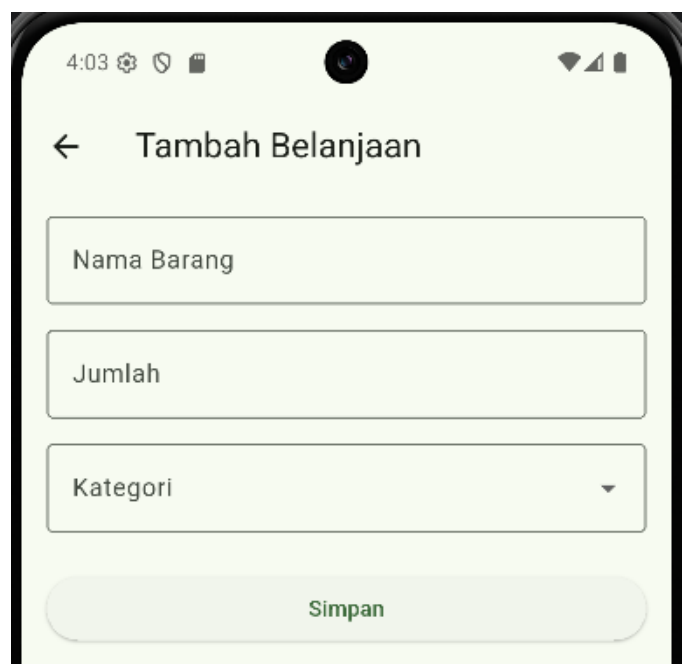
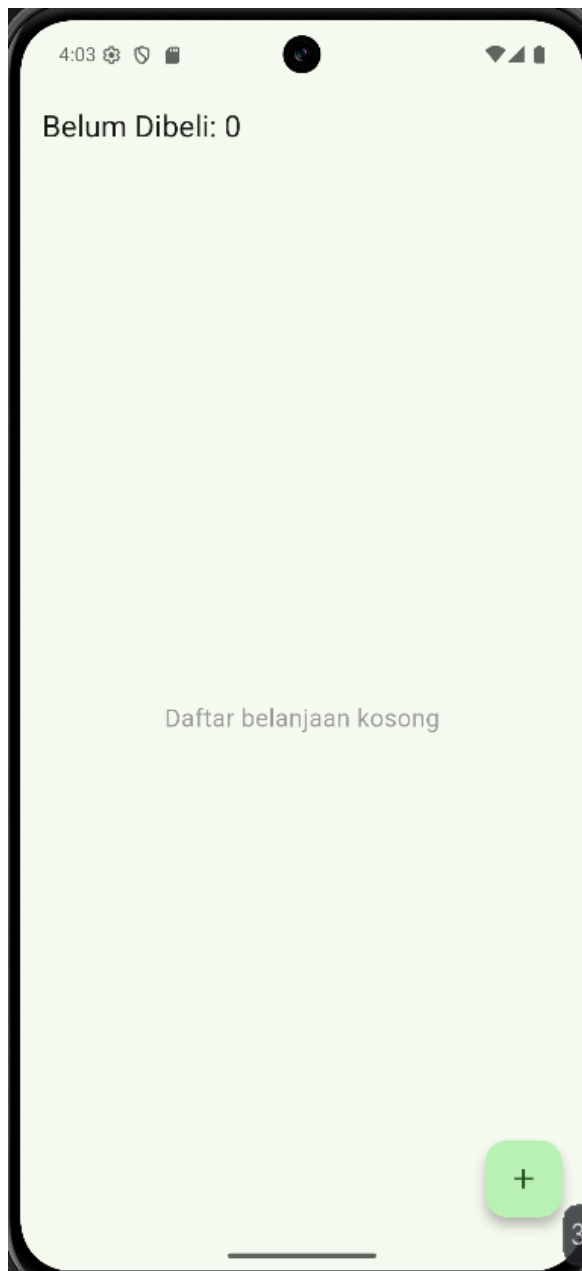
        if (value == null || value.trim().isEmpty) {
            return 'Jumlah wajib diisi';
        }
        final angka = int.tryParse(value.trim());
        if (angka == null || angka <= 0) {
            return 'Jumlah harus berupa angka lebih dari 0';
        }
        return null;
    },
),
const SizedBox(height: 16),

DropdownButtonFormField<String>(
    decoration: const InputDecoration(
        labelText: 'Kategori',
        border: OutlineInputBorder(),
    ),
    items: _daftarKategori
        .map((kategori) => DropdownMenuItem(
            value: kategori,
            child: Text(kategori),
        ))
        .toList(),
    onChanged: (value) {
        setState(() {
            _kategoriTerpilih = value;
        });
    },
    validator: (value) {
        if (value == null || value.isEmpty) {
            return 'Pilih kategori barang';
        }
        return null;
    },
),
const SizedBox(height: 24),

ElevatedButton(
    onPressed: _simpan,
    child: const Text('Simpan'),
),
],
),
),
),
);
}
}

```

Hasil :



Dropdown :



Barang berhasil ditambahkan!

Belum Dibeli: 1



ayam

Jumlah: 2 | Kategori: Makanan &
Minuman



Belum Dibeli: 0



ayam

Jumlah: 2 | Kategori: Makanan &
Minuman

